

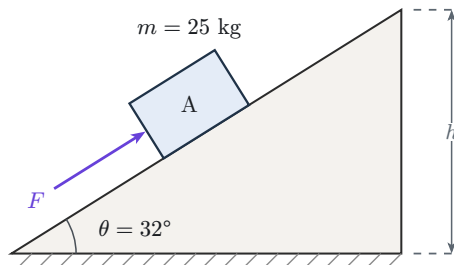
PROBLEM 1

10 marks

Crate on a rough incline

A crate of mass $m = 25 \text{ kg}$ rests on a ramp inclined at $\theta = 32^\circ$ to the horizontal. The coefficients of friction between the crate and the ramp are $\mu_s = 0.45$ and $\mu_k = 0.35$. A force $F = 250 \text{ N}$ is applied to the crate parallel to the ramp, directed up the slope.

- Draw the free-body diagram of the crate and resolve its weight into components along and normal to the ramp.
- Show that the crate cannot remain at rest, and state the direction in which it moves.
- Find the normal force and the acceleration of the crate.
- The crate starts from rest at the foot of the ramp. Find its speed on reaching the top, a height $h = 3.4 \text{ m}$ above the ground.



Take $g = 9.81 \text{ m/s}^2$.

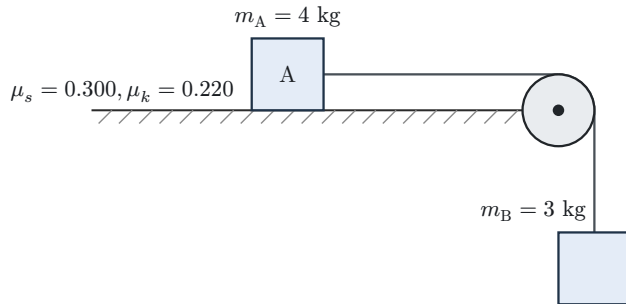
PROBLEM 2

12 marks

Block, cord, and hanging mass

Block A lies on a horizontal table. A light inextensible cord runs from A , passes over a frictionless pulley at the edge of the table, and carries block B . The coefficients of friction between A and the table are given in the figure; the pulley is light and turns freely.

- Draw a separate free-body diagram for each block, and state the relation between their accelerations.
- Determine whether the blocks move when released from rest.
- Find the acceleration of the blocks and the tension in the cord.
- Block B falls a distance $d = 1.2\text{ m}$ to the floor. Find the speed of A at that instant, and the further distance A slides before stopping.



Take $g = 9.81\text{ m/s}^2$.

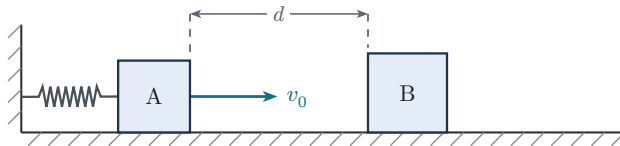
PROBLEM 3

12 marks

Spring launch and collision

Block A , of mass 2.0 kg , is held against a spring of stiffness $k = 800\text{ N/m}$ compressed $x = 0.15\text{ m}$ from its natural length, and is then released. The spring is light and the track is smooth. Block B , of mass 3.0 kg , stands at rest a distance $d = 2.5\text{ m}$ ahead of A .

- Find the speed v_0 with which A leaves the spring.
- The blocks stick together on impact. Find their common velocity and the kinetic energy lost in the collision.
- Find the velocity of each block instead if the collision is perfectly elastic.
- Beyond B the track is rough, with $\mu_k = 0.20$. Taking the collision of part (b), find how far the blocks travel before coming to rest.



Take $g = 9.81\text{ m/s}^2$.

PROBLEM 4

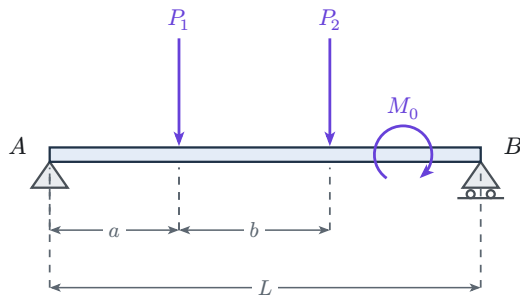
10 marks

Simply supported beam

A light beam AB of length $L = 6.0\text{ m}$ rests on a pin support at A and a roller at B . It carries vertical loads $P_1 = 4.0\text{ kN}$ and $P_2 = 6.0\text{ kN}$ at the distances shown, with $a = 1.8\text{ m}$ and $b = 2.1\text{ m}$, together with a couple $M_0 = 5.0\text{ kN} \cdot \text{m}$.

- Draw the free-body diagram of the beam, showing the reaction components at each support.
- Find the reactions at A and at B .
- Explain why the point of application of M_0 along the beam does not affect either reaction.
- Find the distance from A at which P_2 would have to act for the two reactions to be equal.

Take $g = 9.81\text{ m/s}^2$.



PROBLEM 5

10 marks

Vertical circular loop

A small body of mass $m = 0.40$ kg slides along a smooth horizontal track and runs onto the inside of a vertical circular loop of radius $R = 2.0$ m. Friction is negligible throughout, and the body may be treated as a particle.

- (a) Draw the free-body diagram of the body at the highest point of the loop, and write Newton's second law along the radial direction there.
- (b) Find the least speed v at the highest point for which the body stays in contact with the track.
- (c) Find the speed the body must have on entering the loop at the bottom in order to pass the highest point at that least speed.
- (d) Find the speed at the highest point for which the track pushes on the body with a force equal to the body's weight.

Take $g = 9.81$ m/s².

